DSA06 DATA HANDLING AND VISUALIZATION | Activity: Identify Seasonality Patterns | Slot C

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Time: 20 to 25 minutes Total: 20 marks Tool: R (ggplot2 / base R). Work alone; every student's series is different. Write your answers and paste your plots and code.

Series SP-C15: Retail sales [20 marks]

Scenario: You are the analyst for a retail chain. The series sales (Sales (lakh rupees)) is recorded quarterly. Management believes it follows a repeating pattern and wants it identified before planning. The cycle length is 4 observations. Run the code below to create the series x and answer the tasks.

Evidence: run this code in R (do not edit it).

```
set.seed(6795)
m <- 4                    # observations per cycle (quarterly data)
n <- m * 10               # 10 full cycles
t <- 1:n
trend <- 107.9 + 1.041 * t
sales <- round(trend + 22.2 * cos(2 * pi * (t - 4) / m) + rnorm(n, 0, 6.1), 1)
x <- ts(sales, frequency = m, start = c(2013, 1))
```

Tasks:

1. Look at the series. Plot x (plot(x) or ggplot2). State n, the cycle length, the number of full cycles, and whether the overall trend is rising, falling or flat (fit lm(sales ~ 1:n) and report the slope per quarter). [3]
2. The seasonal pattern. Use decompose(x) and extract $figure (the 4 seasonal values). Name the peak quarter and the trough quarter with their values, and the seasonal swing (peak minus trough). [4]
3. Is it consistent? Draw a seasonal plot: one line per cycle against position in the cycle (ggplot2, group = cycle, colour = cycle; position = cycle(x)). In how many of the 10 cycles is the highest value in the peak quarter from Task 2? Is the pattern consistent? [3]
4. Confirm with autocorrelation. Detrend (d <- resid(lm(sales ~ 1:n))) and compute acf(d, lag.max = 4, plot = FALSE) using observation lags. Report the autocorrelation at lag 4 and at lag 2, and compare with the significance bound 2/sqrt(n). What do the two values say about seasonality? [4]
5. Additive or multiplicative? Compute the range (max minus min) of sales in the first cycle and in the last cycle and their ratio (last / first); decide the type and justify [2]. Remove the seasonal part with decompose(x, type = ...) using your chosen type (subtract for additive, divide for multiplicative), fit lm(adjusted ~ 1:n) and report the slope per quarter [2]. Write one sentence of advice to management based on your findings [2]. [6]

Remember: *seasonality repeats at a fixed, known period; a trend is not a season; if the swings grow with the level, the season is multiplicative.*

Code of Conduct: I certify that this work is my own. Signature: _____

Activity -4 (8)
(7)

Fitness Center case

```
set.seed(4915)
n <- 14
outdoor_temp <- round(rnorm(n, 35.1, 5.8), 1)
• gym_visits <- round(257 + 59.9 * (0.91 * (outdoor_
  temp - 35.1) / 5.8 + 0.41 * rnorm(n)), 0)
• shakes_sold <- round(70 + 17.4 * (-0.87 * (outdoor
-temp - 35.1) /5.8 + 0.49 * rnorm(n)), 0)
• load_dev <- round(seq(-16.0, 16.0, length.out = n)
                [order(runif(n)], 1)
• injuries <- round(2.3 + 8.5 * (load_dev/ 16.0)^2 +
                rnorm(n, 0, 0.3), 0)
• trainers <- round(rnorm(n, 7, 1.7), 0)
• towel_loads <- round(21 + 5.8 * (0.18 * (trainers -7)
/1.7 + 0.98 * rnorm(n)), 0)
• trainers[10] <- 13
• towel_loads[10] <- 39
• case <- data.frame(outdoor_temp, gym_visits,
shakes_sold, load_dev, injuries, trainers, towel_loads)

Task-1
round(cor(case), 2)
nrow(case)
choose(ncol(case), 2)

# correlogram
library(ggplot2)
cor_data <- as.data.frame(as.table(cor(case)))
names(core_data) <- c("variable", "variable2", "correlation")
```

```
ggplot(cor_data, aes(variable, variable2, fill = correlation)),
  geom_ title (color = "white") +
  geom_ text (aes (lable = sprintf ("%.2f", correlation)),
  size = 3.5) +
  scale_fill_ gradient2(low = "blue", mid = "white", high = "red",
    midpoint = 0) +
  labs( title = "Fitness Centre correlation Matrix",
        x = "", y = "") +
  theme_ minimal() +
  theme (axis. text. x = element_ text (angle = 45, hjust = 1))
# Task 2
  cm <- cor(case)
  cm [lower.tri(cm, diag = TRUE)] <- NA
  sort (cm, decreasing = TRUE, na.last = NA)
  sort (cm, na.last = NA)
# Task 3
Plot (injuries ~ load_dev, data = case, pch = 19,
    main = "Injuries vs Training Load Deviation",
    xlab = "Training Load Deviation", ylab = "Injuri
abline (lm(injuries ~ load_dev, data = case), col = "blue",
                                    lwd = 2)
quad_fit <- lm (injuries ~ load_dev + I(load_dev^2
dat = case)
  ord <- order (case $ load_dev)
  summary (quad_fit) $r.squared.
# Task 4
  Plot (towel_ loads ~ trainers, data = case, pch =
    main = "Towel Loads vs Trainers on Duty",
    xlab = "Trainers on Duty", ylab = "Towel Laun
                                    Loads")
```

```
text(case$trainers, case$towel_loads, labels = seq_len
                (nrow(case)), pos=3, cex=8)
cor(case$trainers, case$towel_loads)
cor(case$trainers[-10], case$towel_loads[-10])

# Task 5
    cor(case$gym_visits, case$shakes_sold)
    cor(case$outdoor_temp, case$gym_visits)
    cor(case$outdoor_temp, case$gym_sold)
    cor(resid(lm(gym_visits ~ outdoor_temp, data = case
        resid(lm(shakes_sold ~ outdoor_temp, data = case))"
```